Overview of the Nervous System

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Learning Objectives (2026)

1. Describe the structural and functional organization of the nervous system, including the CNS, PNS, sensory pathways, motor pathways, and autonomic divisions. (CLO 1, CLO 2)
2. Apply standard neuroanatomical terminology to describe orientation, sections, gray matter, white matter, nuclei, tracts, nerves, and ganglia. (CLO 1, CLO 5)
3. Relate the structural organization of neurons, glial cells, and major neural pathways to normal nervous system function and the principles of neurological localization. (CLO 1, CLO 2, CLO 5)

Pre-Session Assignment

- Preview the PowerPoint slides. You are not expected to memorize details before class.
- Watch: B&B: Cells of the Nervous System (23 min)

Structural Division – Where Structures Are Located

Central Nervous System (CNS)

- Brain
- Spinal cord

Peripheral Nervous System (PNS)

- Cranial nerves
- Spinal nerves
- Peripheral nerves
- Ganglia

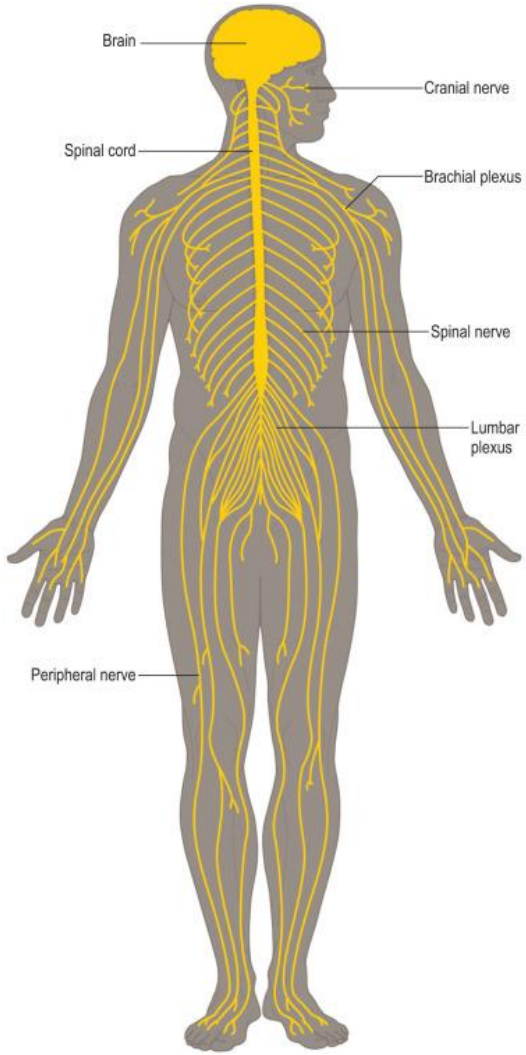
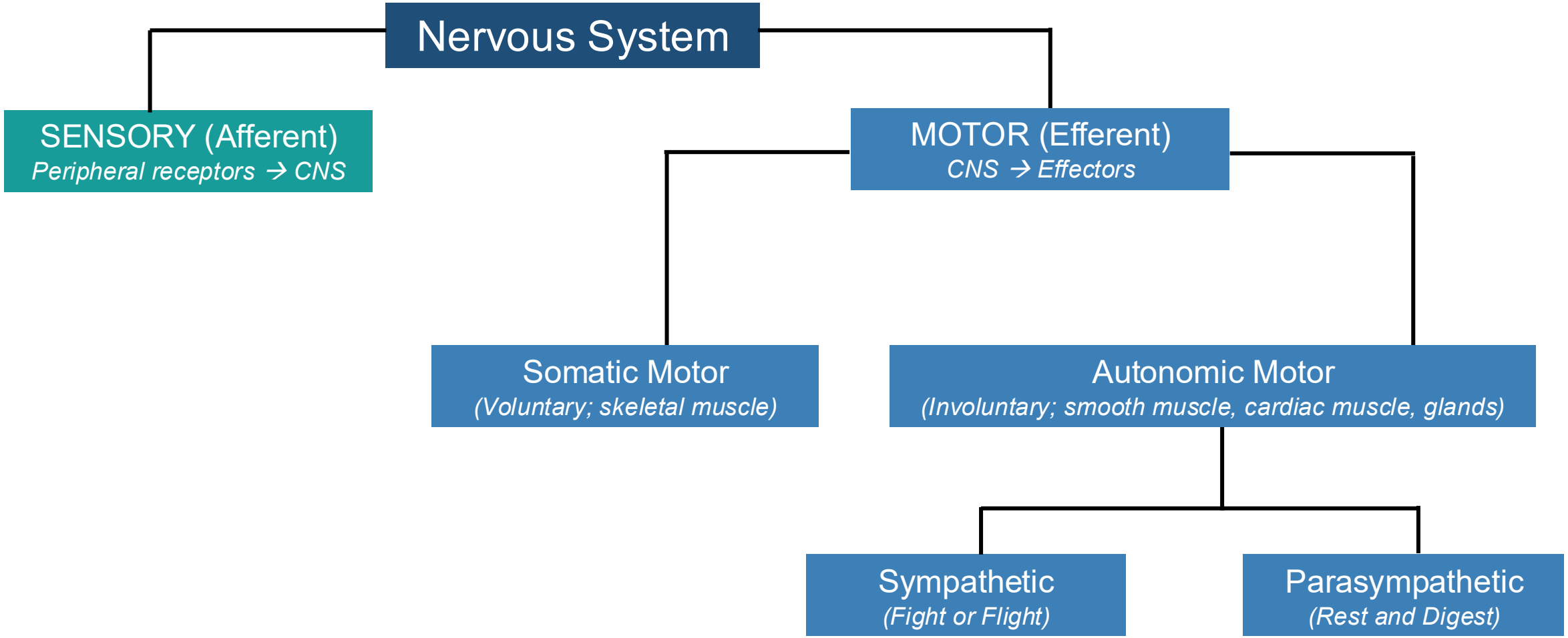


Fig. 1.5 Central and Peripheral Nervous System.
Crossman, Neuroanatomy, 6e.

Functional Organization of the Nervous System



Structural divisions tell us where neural structures are located.
Functional divisions describe the direction and function of neural pathways.

Spinal Nerve

Anterior Root

- Carries motor (efferent) fibers
- Somatic motor: cell bodies in anterior horn
- Autonomic motor: preganglionic neuron cell bodies in intermediate gray matter

Posterior Root

- Carries sensory (afferent) fibers
- Sensory neuron cell bodies in the dorsal root ganglion

- Anterior + posterior roots unite → **mixed spinal nerve**
- Spinal nerve exits via intervertebral foramen
- Divides into mixed anterior and posterior rami

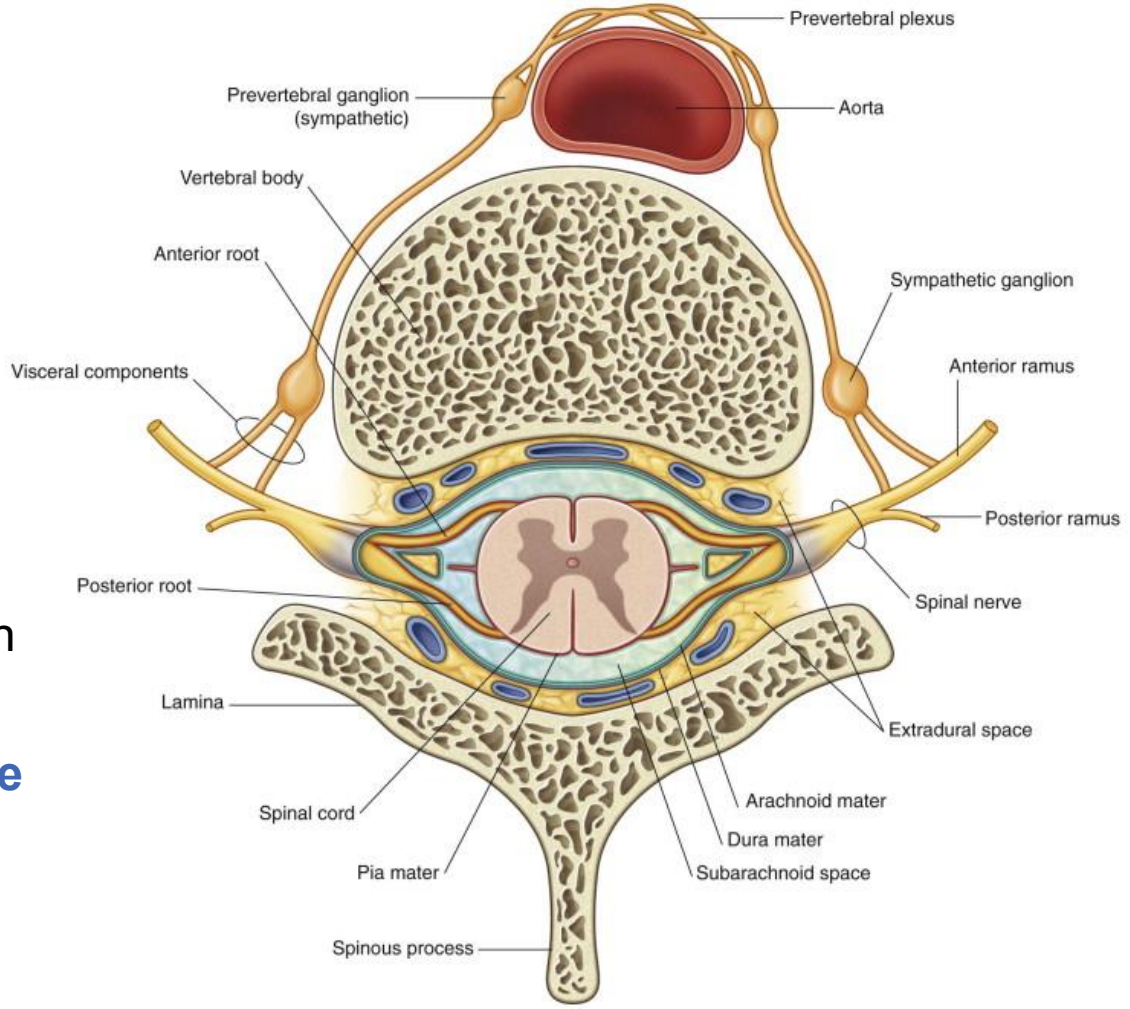


Fig. 2.9 Spinal nerves (transverse section).
Anatomy for Students, 5e.

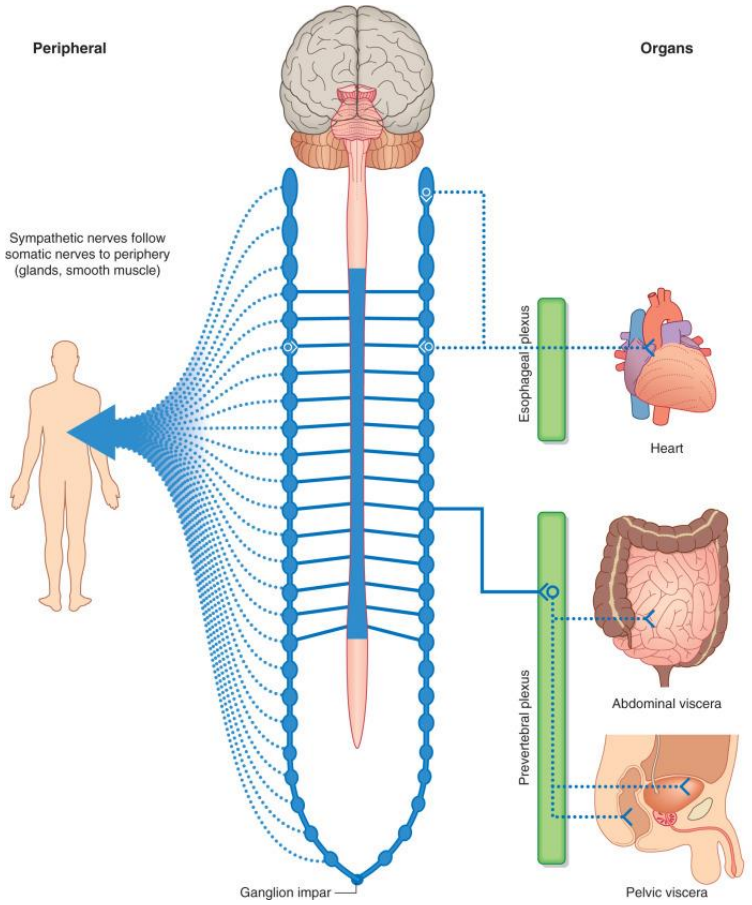
Autonomic Nervous System

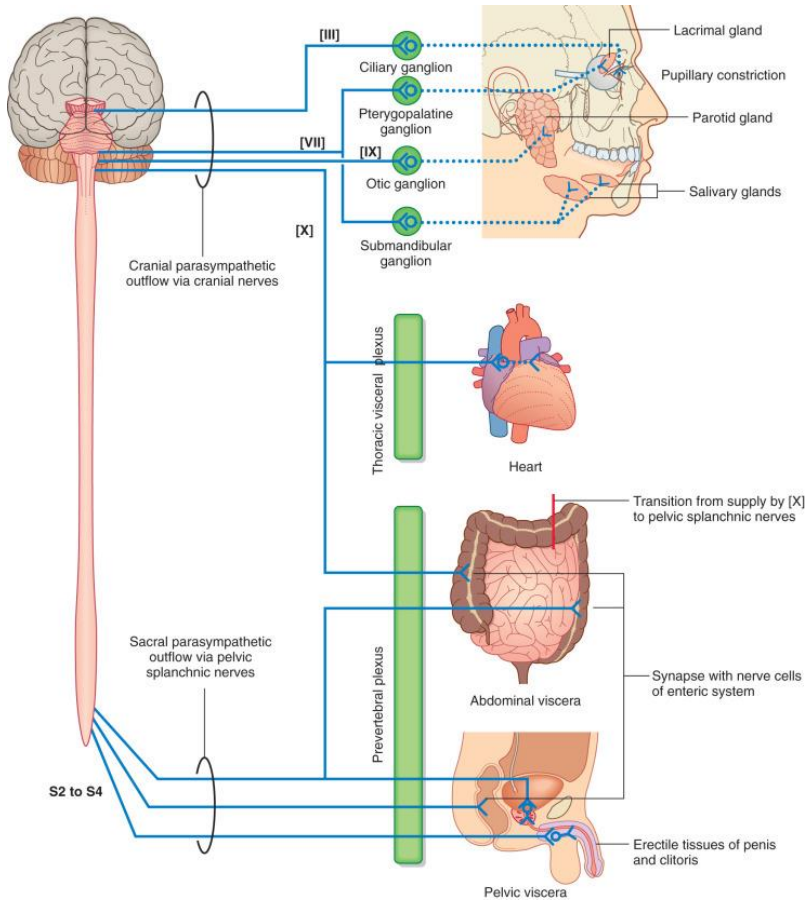
Autonomic motor pathways regulate:

- Smooth muscle
- Cardiac muscle
- Glands

Major divisions:

- Sympathetic → thoracolumbar outflow
- Parasympathetic → craniosacral outflow





Both divisions generally use a **two-neuron motor pathway** from the CNS to the target organ

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Fig. 1.42, 1.47 GAS, 4e.

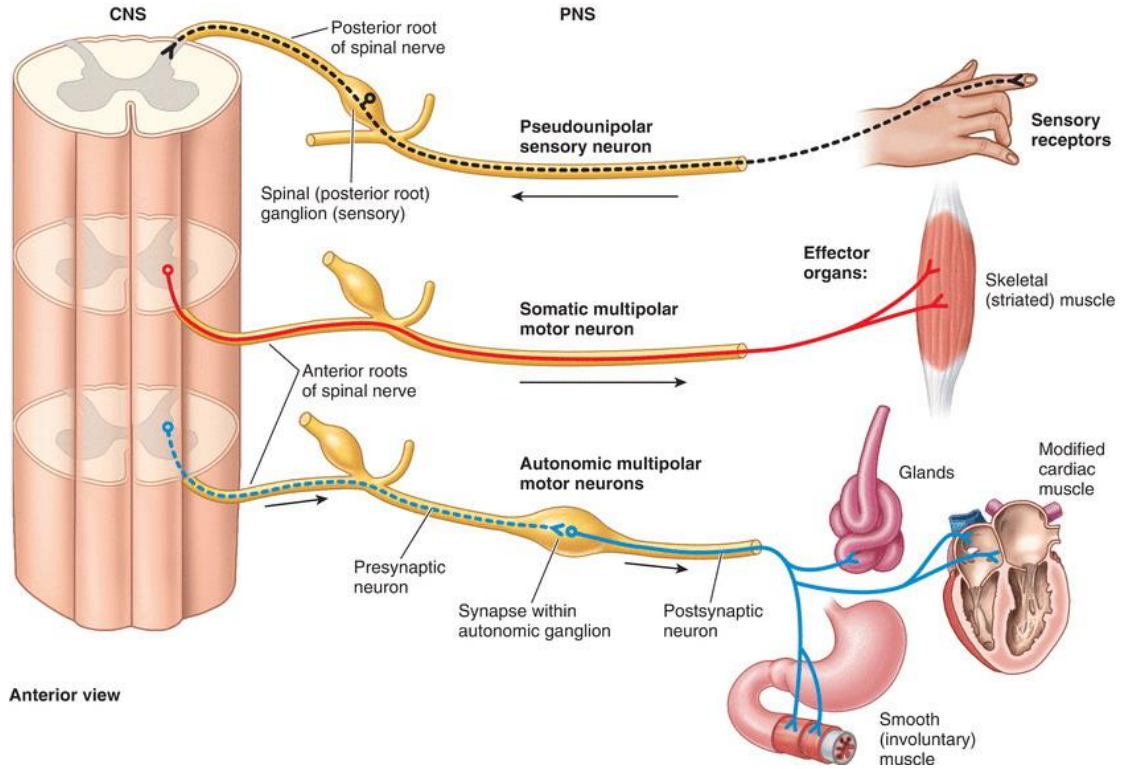
Somatic vs Autonomic Motor Pathways

Somatic Motor

- Single motor neuron
- CNS → skeletal muscle

Autonomic Motor

- Two-neuron pathway
- Preganglionic neuron: CNS → autonomic ganglion
- Postganglionic neuron: autonomic ganglion → target organ
- Targets: smooth muscle, cardiac muscle, glands



Autonomic motor pathways
require a synapse in an autonomic ganglion

Fig. 1.42 Clinical Oriented Anatomy, 8e.

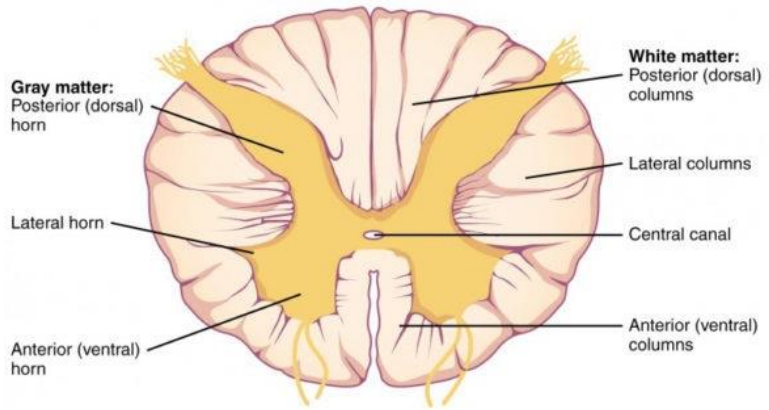
Sympathetic System: Thoracolumbar Outflow

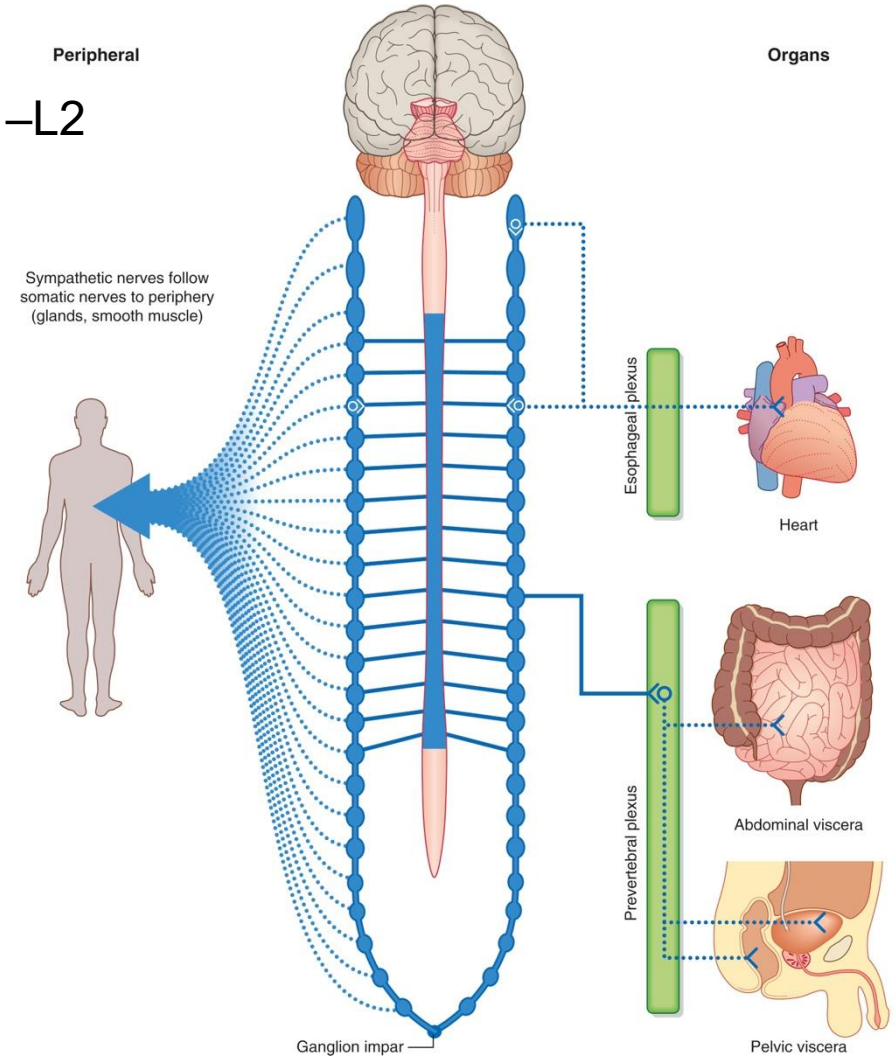
Preganglionic neurons

- Cell bodies in the lateral horn (intermediolateral cell column), T1–L2
- Axons exit through ventral roots → enter spinal nerves
- Enter the sympathetic trunk through white rami communicantes

Once in the sympathetic trunk, preganglionic fibers may:

- Synapse at the same level
- Ascend or descend before synapsing
- Pass through without synapsing → splanchnic nerves



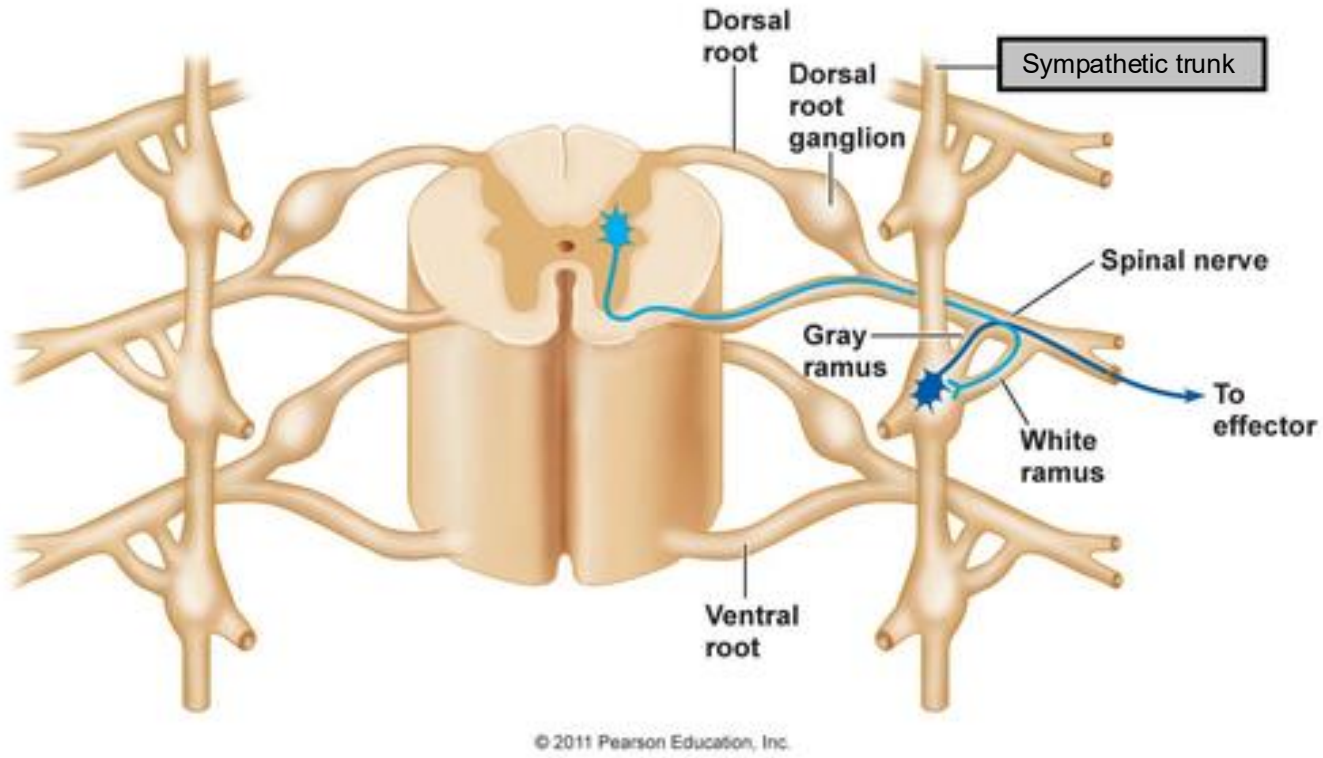


Thoracic spinal cord. Courtesy of Univ of Michigan Med School, ©2012; Fig 1.44 GAS, 5e.

White and Gray Rami Communicantes

- White rami: carry preganglionic sympathetic fibers into the sympathetic trunk
- Gray rami: carry postganglionic sympathetic fibers back to the spinal nerve
- White rami: T1–L2 only
- Gray rami: present at all spinal levels

Gray rami allow sympathetic fibers to reach the body wall and limbs at all spinal levels



Sympathetic Splanchnic Nerves

Thoracic and Lumbar Splanchnic Nerves

- Carry primarily preganglionic sympathetic fibers
- Fibers pass through the sympathetic trunk without synapsing
- Travel to prevertebral ganglia and autonomic plexuses

Sacral Splanchnic Nerves

- Preganglionic fibers descend in the sympathetic trunk
- Synapse in sacral sympathetic ganglia
- Postganglionic fibers leave as sacral splanchnic nerves
- Contribute to the inferior hypogastric plexus

Do not confuse:

- Sacral splanchnic nerves = sympathetic
- Pelvic splanchnic nerves (S2–S4) = parasympathetic

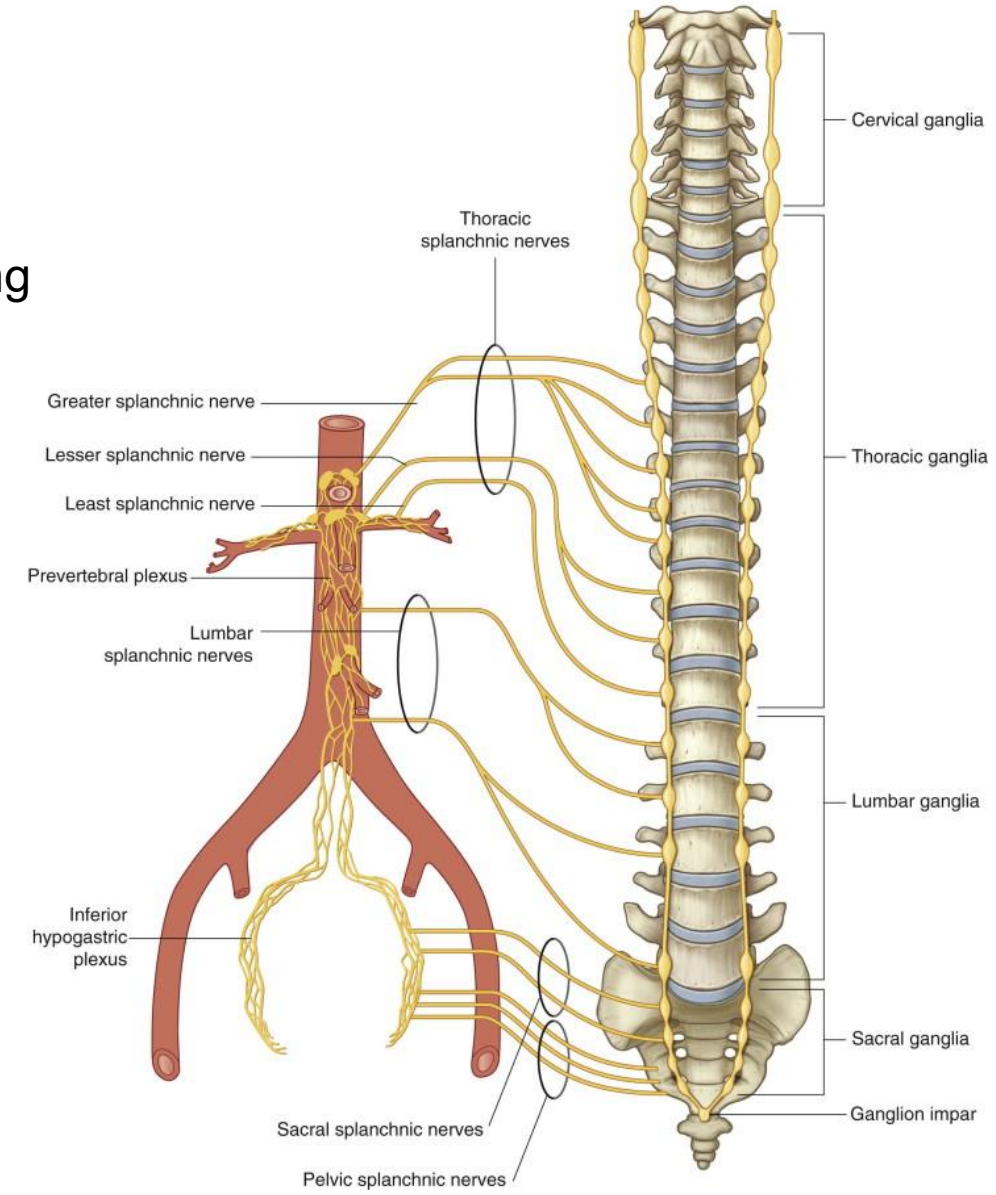


Fig. 4.136 Splanchnic nerves. GAS, 4e.

Sympathetic Innervation of Abdominal Viscera

Sympathetic Pathway

- Preganglionic fibers pass through the sympathetic trunk without synapsing
- Form thoracic and lumbar splanchnic nerves
- Synapse primarily in prevertebral ganglia
- Postganglionic fibers follow arteries to abdominal viscera

Major Prevertebral Ganglia / Plexuses

- Celiac → primarily foregut
- Superior mesenteric → primarily midgut
- Inferior mesenteric → primarily hindgut

Spinal cord → ventral root → spinal nerve → white ramus → sympathetic trunk → thoracic/lumbar splanchnic nerve → prevertebral ganglion → periarterial plexus → organ

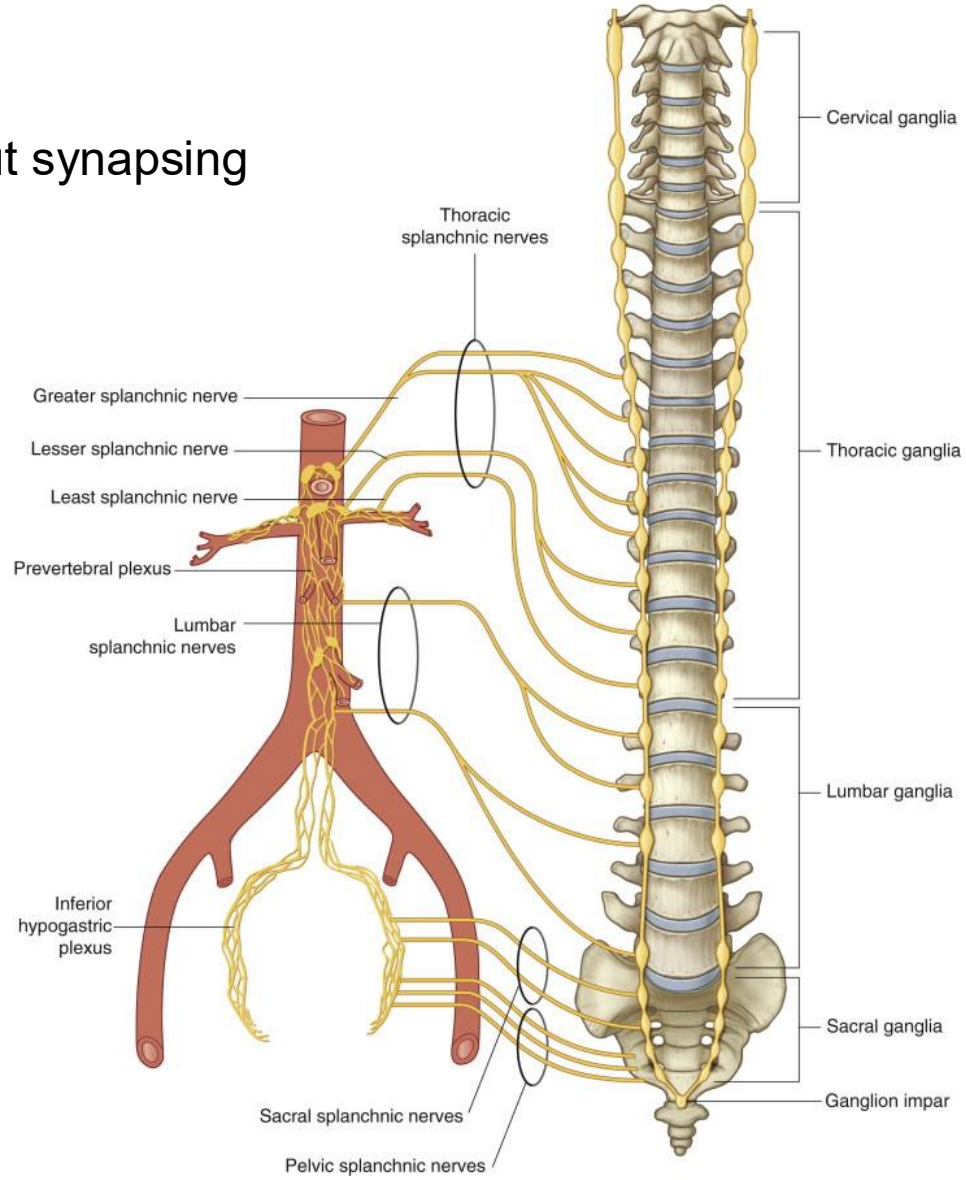


Fig. 4.136 Splanchnic nerves. GAS, 4e.

Parasympathetic System: Craniosacral Outflow

Preganglionic neurons originate in:

- Brainstem → CN III, VII, IX, X
- Sacral parasympathetic nucleus (S2–S4)

Organization

- Long preganglionic fibers
- Synapse in ganglia near or within target organs
- Short postganglionic fibers → targets

Major Visceral Pathways

- CN III, VII, IX → head and neck structures
- Vagus nerve (CN X) → thoracic viscera, foregut, and midgut
- Preganglionic fibers from S2–S4 form pelvic splanchnic nerves → hindgut and pelvic viscera

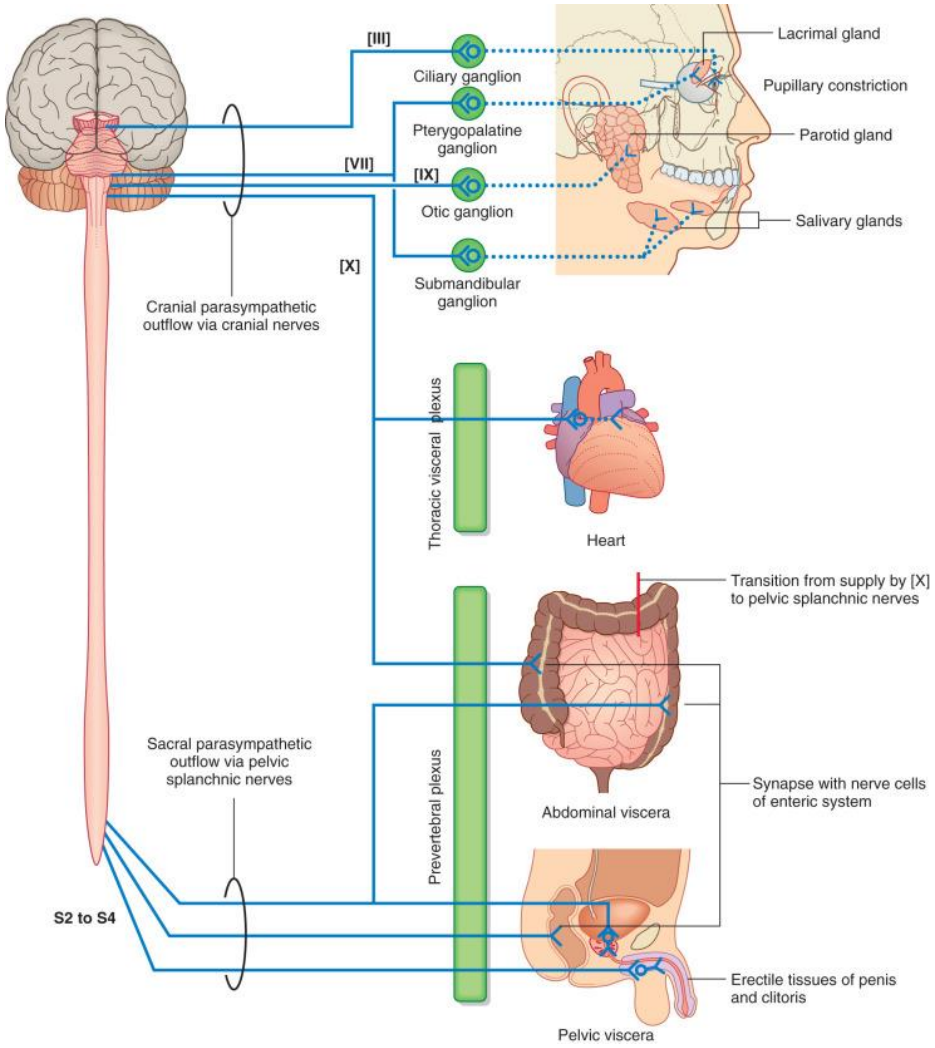


Fig. 1.47 GAS, 4e.

Parasympathetic Innervation of the Abdominal Viscera

Parasympathetic Pathway

- Vagus nerve (CN X) → foregut and midgut
- Pelvic splanchnic nerves (S2–S4) → hindgut
- Preganglionic fibers extend to the target organ
- Synapse in terminal/intramural ganglia

Functional Effects

- ↑ GI motility
- ↑ GI secretion

Vagus nerve or pelvic splanchnic (S2–S4) nerves → terminal/intramural ganglia → short postganglionic fibers → GI effectors

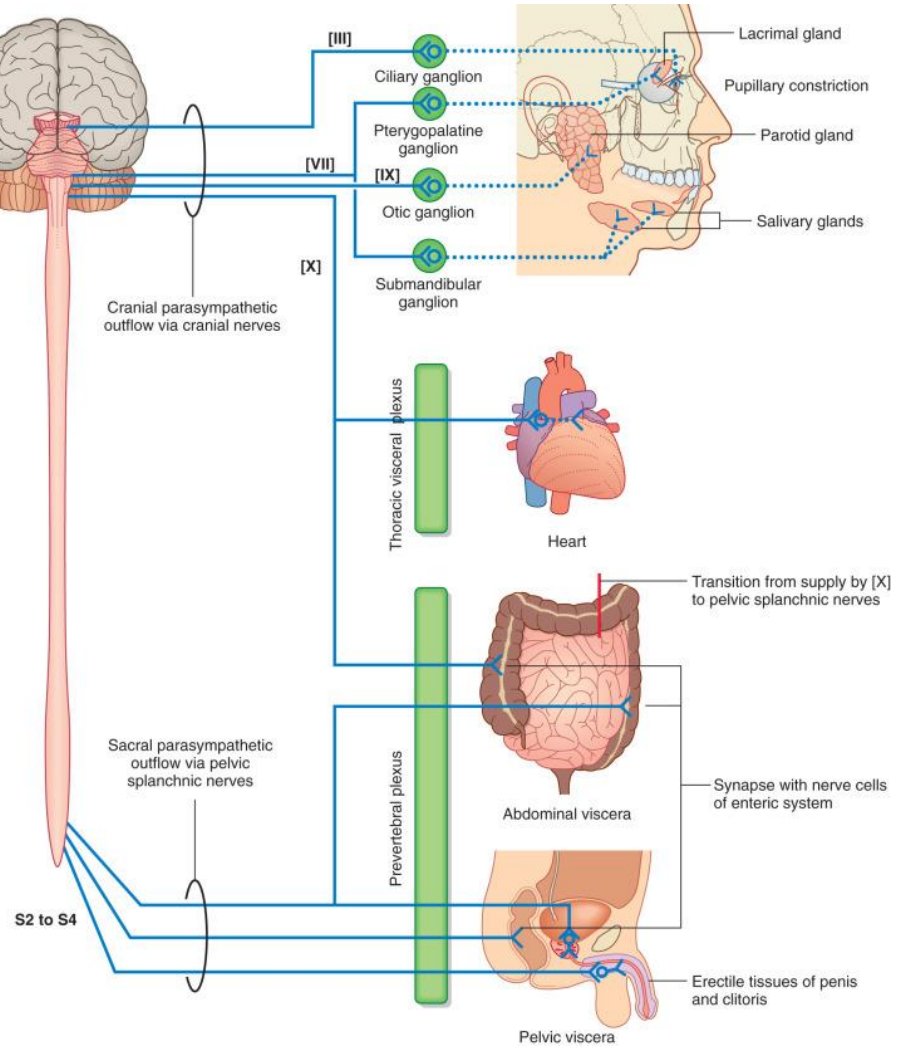


Fig. 1.47 GAS, 4e.

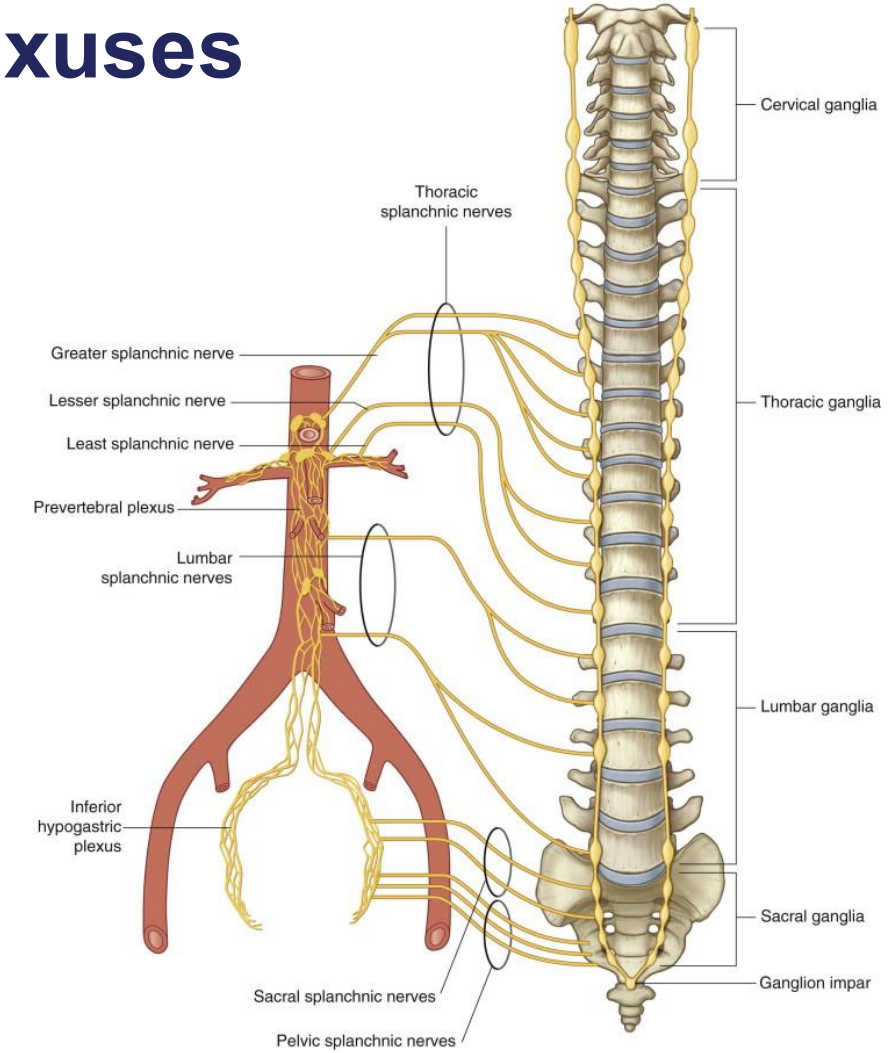
Pelvic Autonomics: Hypogastric Plexuses

Sympathetic

- Abdominal aortic plexus → superior hypogastric plexus
- Superior hypogastric plexus → right and left hypogastric nerves
- Hypogastric nerves → inferior hypogastric plexuses
- Sacral splanchnic nerves contribute to inferior hypogastric plexus

Parasympathetic

- Pelvic splanchnic nerves (S2–S4)
- Contribute to the inferior hypogastric plexus



- Inferior hypogastric plexus contains both sympathetic and parasympathetic fibers
- Distributes to pelvic viscera and hindgut

Fig. 4.136 Splanchnic nerves. GAS, 4e.

Enteric Nervous System

Intrinsic Neural Network of the GI tract

- Myenteric (Auerbach) plexus → primarily regulates motility
- Submucosal (Meissner) plexus → secretion and local blood flow
- Can coordinate GI activity independently
- Modulated by sympathetic and parasympathetic input

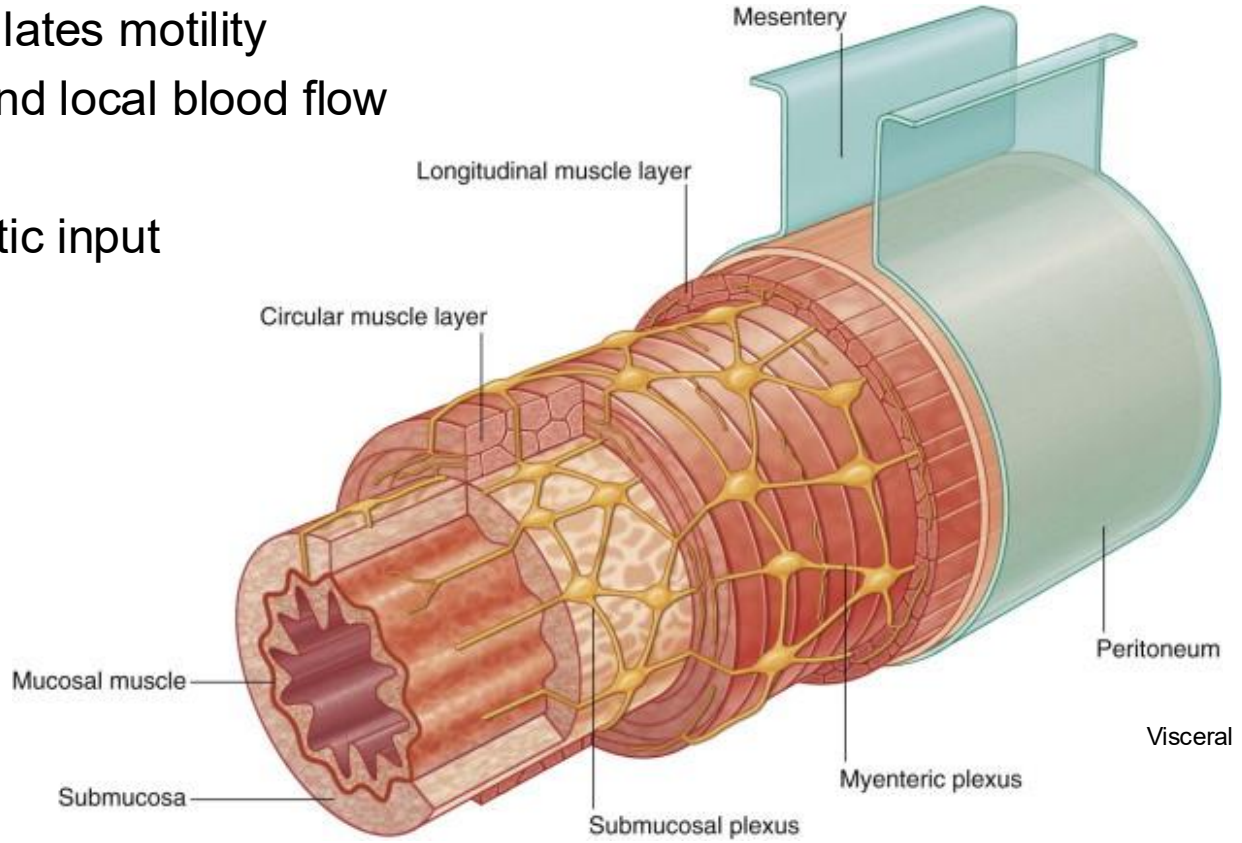


Fig. 4.139 The enteric system. GAS, 4e.

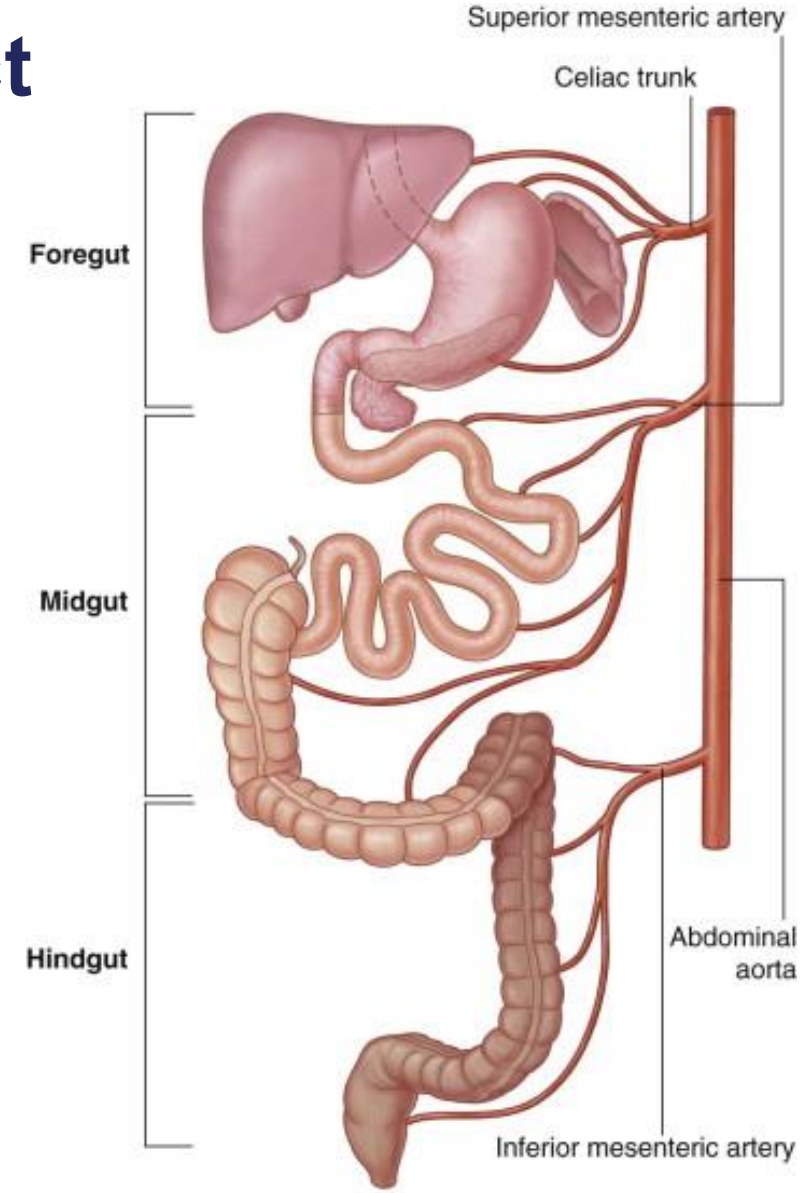
Autonomic Innervation of the GI Tract

Sympathetic

- Foregut & midgut → thoracic splanchnic nerves
- Hindgut → primarily lumbar splanchnic nerves
- Synapse in prevertebral ganglia
- Postganglionic fibers follow periarterial plexuses
- Generally, ↓ motility and secretion

Parasympathetic

- Foregut & midgut → vagus nerve (CN X)
- Hindgut → pelvic splanchnic nerves (S2–S4)
- Synapse in terminal/intramural ganglia
- Generally, ↑ motility and secretion



Somatic vs Visceral Sensory Pathways

Somatic Sensory (Afferent)

- From skin, skeletal muscle, and joints
- Carries touch, pain, temperature, and proprioception
- Cell bodies in dorsal root ganglia (DRG)
- Fibers enter the spinal cord through posterior roots

Visceral Sensory (Afferent)

- From viscera and blood vessels
- Carries distension, chemical changes, and visceral pain
- Cell bodies in sensory ganglia (DRG, cranial sensory ganglia)
- Spinal visceral afferents enter through posterior roots

Sensory → CNS: Somatic afferents | Visceral afferents

Motor → Effectors: Somatic → skeletal muscle | Autonomic → viscera

Clinical Anatomy Pearl

- Visceral pain afferents → commonly accompany sympathetic pathways
- Physiologic/reflex afferents → commonly accompany parasympathetic pathways

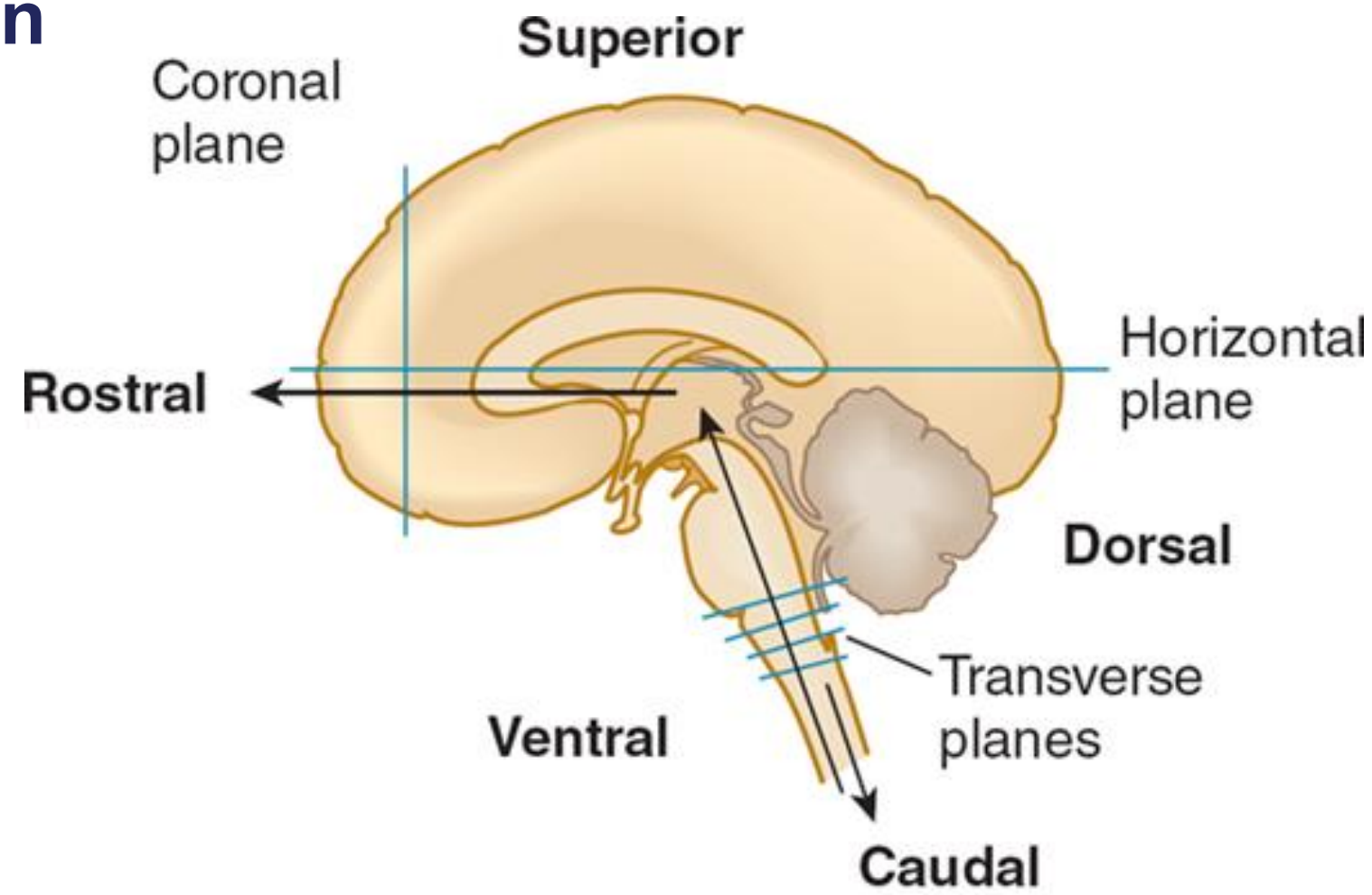
Terms of Orientation

Spinal Cord

- Anterior = Ventral
- Posterior = Dorsal
- Superior = Rostral
- Inferior = Caudal

Brain

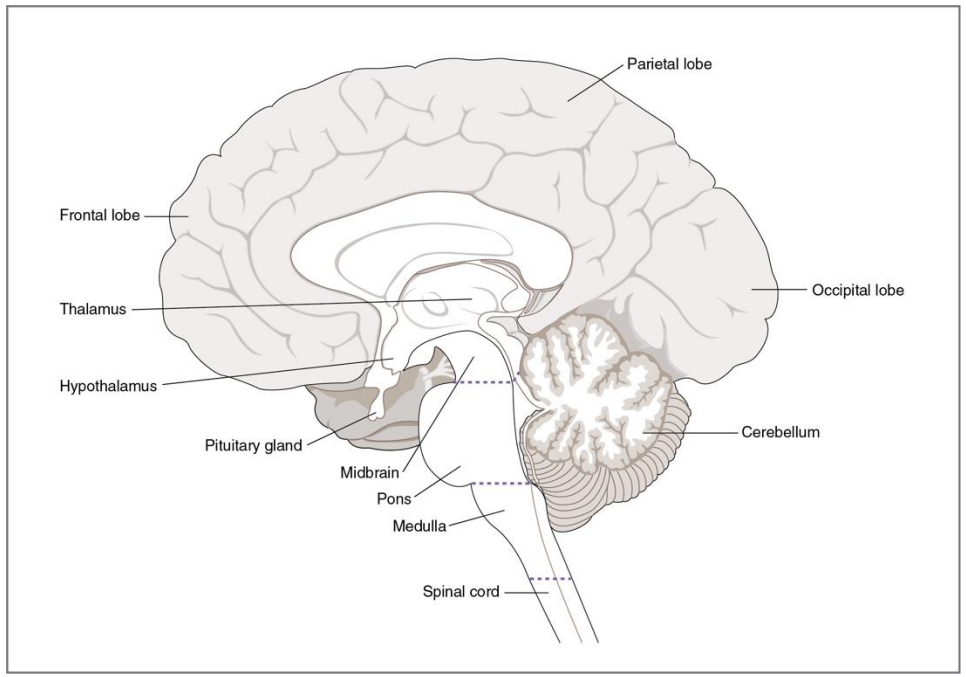
- Anterior = Rostral
- Posterior = Caudal
- Superior = Dorsal
- Inferior = Ventral

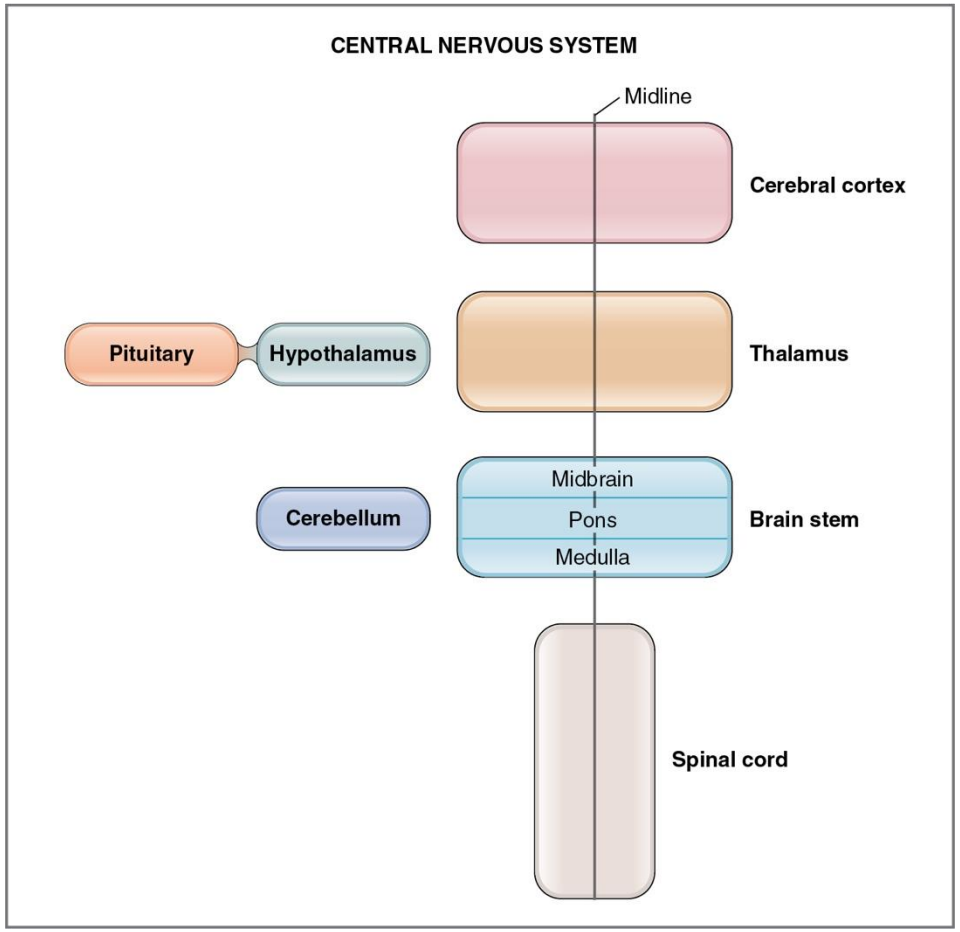


Source: Stephen G. Waxman: *Clinical Neuroanatomy, 30th Edition*
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Orientation shifts at the cephalic flexure

Central Nervous System Overview





Brain

- Cerebrum
- Brainstem
- Cerebellum

Spinal cord

Fig. 3.1, 3.2 Costanzo, 7e.

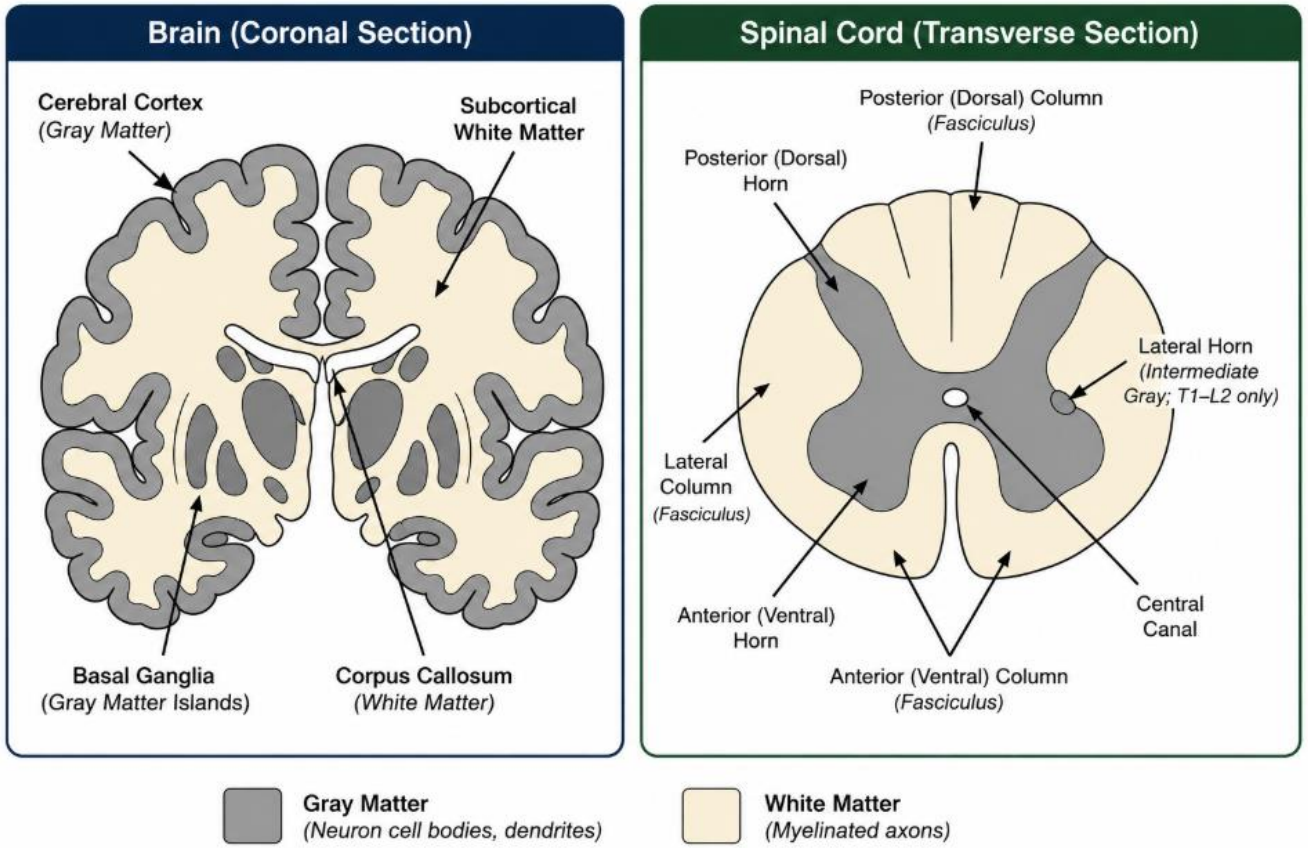
Gray Matter vs. White Matter

Gray Matter

- Neuronal cell bodies, dendrites, and unmyelinated axons
- Site of information processing
- Appears gray in fresh tissue

White Matter

- Myelinated axons (tracts)
- Carries information between regions
- Appears white due to myelin



Organization

- Brain: Gray matter outside, white matter inside
- Spinal Cord: Gray matter inside, white matter outside

Gray matter lesions: focal deficits (e.g., stroke → weakness, aphasia)
White matter lesions: tract conduction deficits (multiple sclerosis → widespread or dissociated deficits

Neuron

Three Main Parts

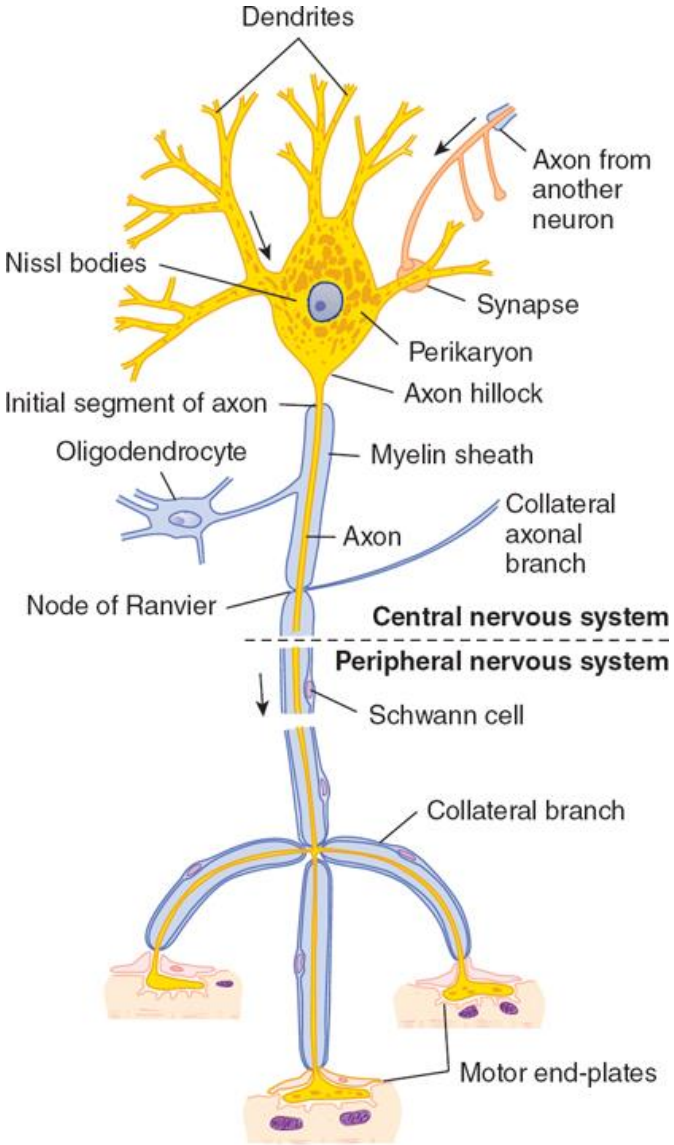
- Dendrites → receive input
- Cell body (soma) → integrates signals
- Axon → transmits output

Signal Transmission

- Action potentials begin at the axon hillock
- Axon terminals release neurotransmitters at synapses

Organization

- Tracts = axon bundles in the CNS
- Nerves = axon bundles in the PNS
- Nuclei = neuron cell body clusters in the CNS
- Ganglia = neuron cell body clusters in the PNS



Source: Stephen G. Waxman: *Clinical Neuroanatomy, 30th Edition*
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Neuronal loss is largely irreversible
Early reperfusion preserves brain function

Neuroglia (Glial Cells)

Major Functions

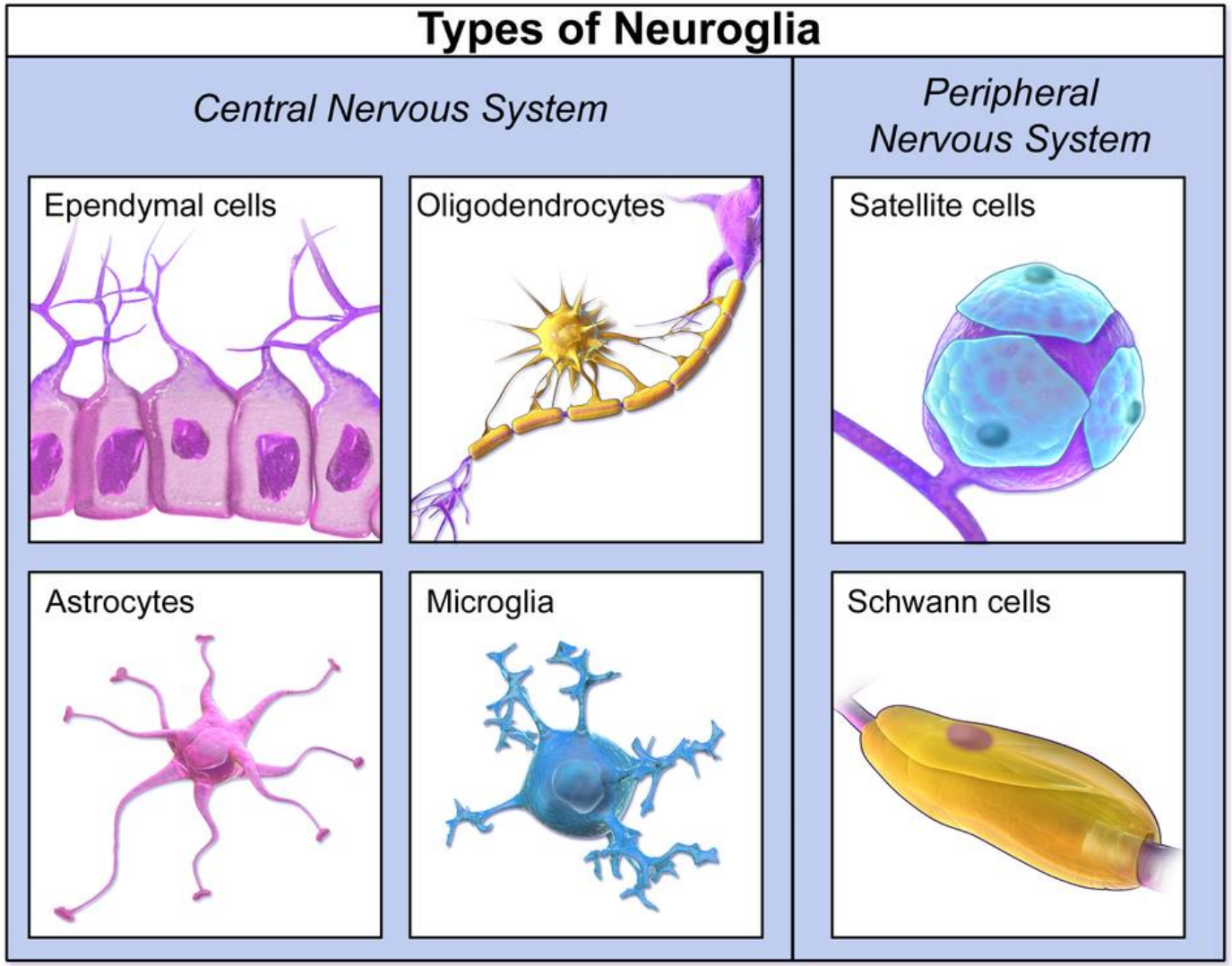
- Support and protect neurons
- Form myelin
- Participate in immune defense

CNS Glia

- Astrocytes
- Oligodendrocytes
- Microglia
- Ependymal cells

PNS Glia

- Schwann cells
- Satellite cells



Medical gallery of Blausen Medical 2014. www.Blausen.com

High-Yield Glial Cells

Cell Type	Primary Function	Clinical Association
Astrocyte	Blood-brain barrier, support	Astrocytoma; Glioblastoma (GBM)
Oligodendrocyte	CNS myelin	Multiple sclerosis
Microglia	CNS macrophages (immune defense)	Neuroinflammation
Ependymal Cell	CSF production / circulation	Ependymoma
Schwann Cell	PNS myelin, axonal regeneration	Guillain-Barré, Schwannoma
Satellite Cell	Support neurons in ganglia	Herpes zoster

Anatomy Pearls

- Oligodendrocyte = CNS myelin
- Schwann cell = PNS myelin
- One oligodendrocyte → many axons
- One Schwann cell → one axon segment

Clinical Pearl

Optic Nerve ≠ Typical Peripheral Nerve

- CNS tract
- Myelinated by oligodendrocytes
- Optic neuritis is common in MS

Summary

- Nervous system organization includes CNS/PNS and sensory/motor divisions.
- Anterior roots are motor; posterior roots are sensory; spinal nerves and rami are mixed.
- Autonomic motor pathways use preganglionic and postganglionic neurons.
- Sympathetic: thoracolumbar (T1–L2); parasympathetic: craniosacral.
- Sympathetic fibers travel through sympathetic trunks, splanchnic nerves, and autonomic plexuses.
- Gray matter contains neuronal cell bodies; white matter contains predominantly myelinated axons.
- Neurons transmit signals; glial cells support, protect, and myelinate neurons.

Overview of the Nervous System

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Schedule an Appointment

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